

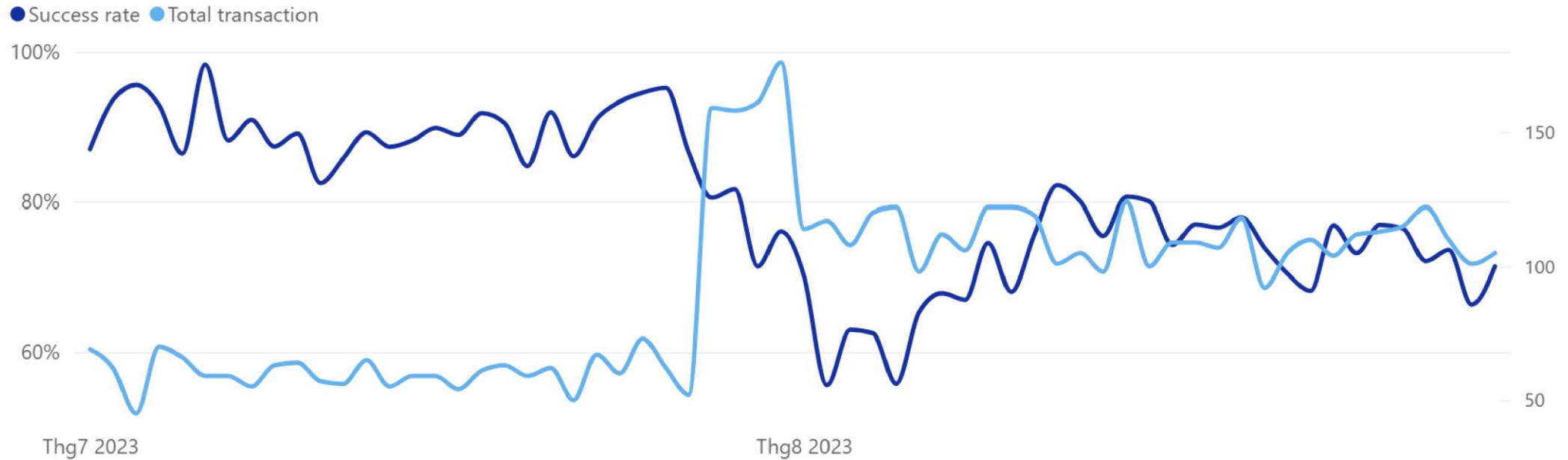
ROOT CAUSE ANALYSIS

# Payment Success Rate Drop Analysis

Performance Evaluation & Strategic Diagnostics — Q3/2023

## Success rate dropped suddenly and sharply starting from July 28th

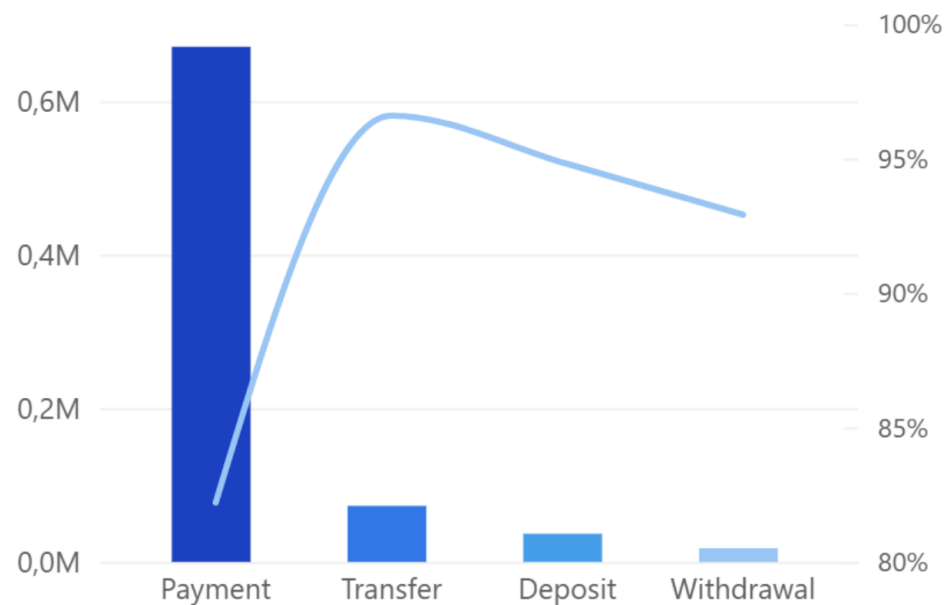
Success rate and Total transaction by Date



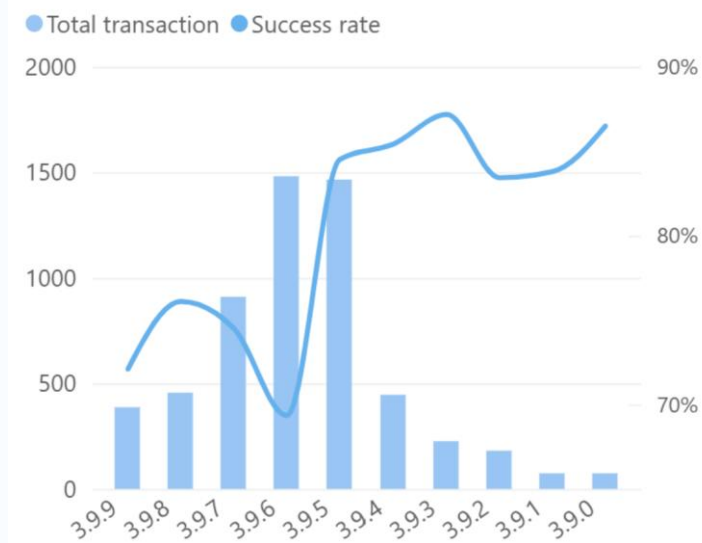
On July 26–27, the success rate remained stable at 92–93%. Immediately following July 28, it plummeted straight down to 82% and has not recovered to its previous benchmark level since.

## Payment success rate suffered the largest decline; current versions unoptimized

### Total revenue and Success rate



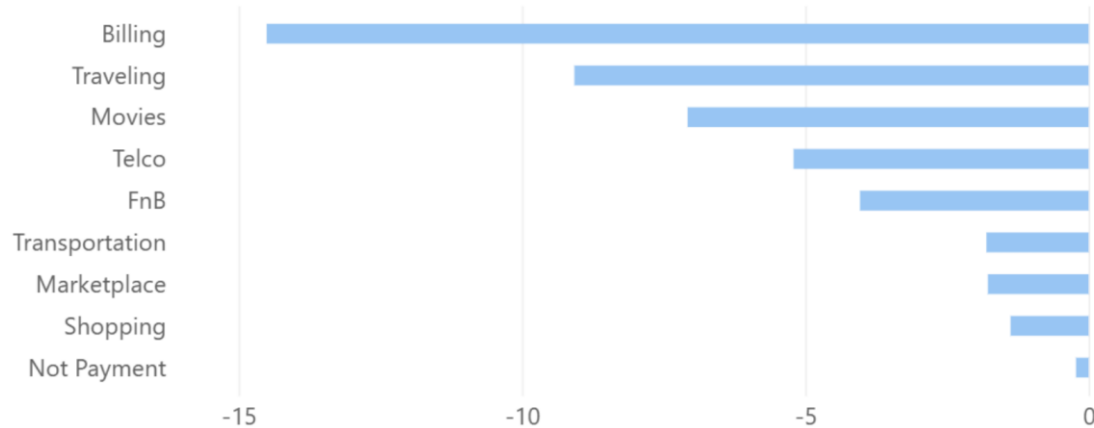
### Total transaction and Success rate by app\_version



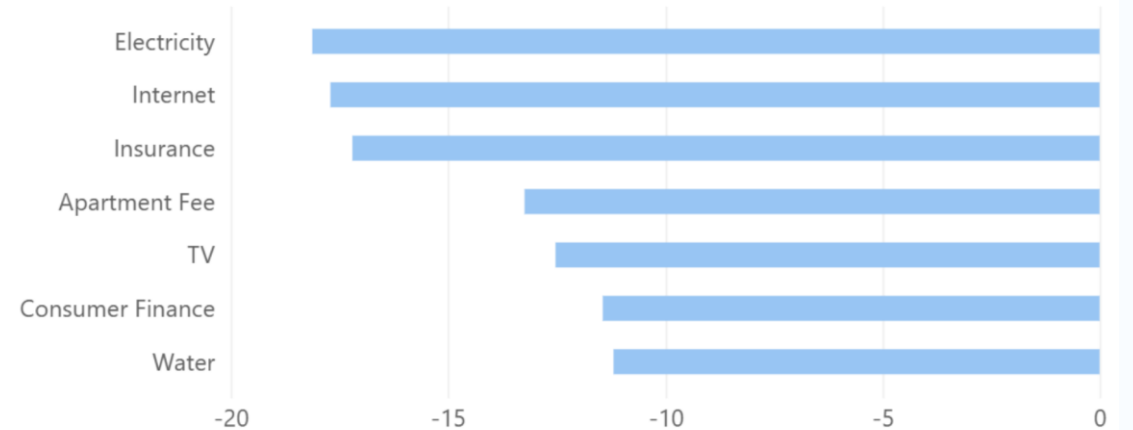
This drop occurred predominantly in the Payment segment, the company's highest revenue generator. Subsequent bug-fix releases for version 3.9.6 have failed to improve the overall metrics.

## Billing success rate suffered a severe and systemic decline

Gap success rate btw moth 7 and 8 in payment



Gap success rate btw moth 7 and 8 in billing



Billing experienced the most severe impact with a deep reduction of -14.5 pp, far exceeding other categories like Traveling or Movies. This widespread failure rules out single partner/biller faults, pointing directly to core processing workflows and Payment Gateway API issues.

## User segmentation factors do not influence the failure rate

### Age Group

Drop rates fluctuate consistently between -14.2% and -15.2% across all age demographics (18–24, 25–34, 35–44, 45+), showing no age group immunity.

### Loyalty Tier

Both new customers (Bronze: -14.6%) and VIP tier customers (Platinum: -16.7%) encountered identical errors, eliminating new user onboarding hypotheses.

### Network Connection

Users with strong connections (Excellent: -14.8%) or poor connections (Poor: -15.0%) failed equally, confirming this is not an ISP or network carrier issue.

### Activity Level

Both high activity users (High: -17.7%) and low activity users (Low: -11.9%) saw steep declines, confirming the core issue lies deep within application logic.

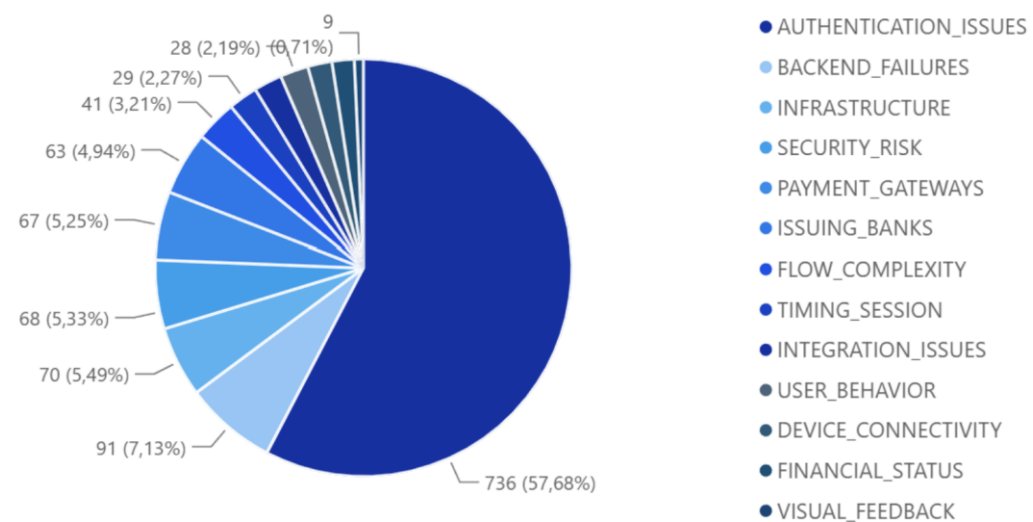
## Payment methods are not the primary bottleneck causing errors

| sub_category     | credit_card | debit_card | linked_bank | local_card | wallet_balance |
|------------------|-------------|------------|-------------|------------|----------------|
| Apartment Fee    | 23,50%      | 17,49%     | 19,69%      |            | 22,97%         |
| Consumer Finance | 23,66%      | 19,28%     | 22,78%      | 33,33%     | 21,15%         |
| Electricity      | 20,32%      | 21,12%     | 22,97%      | 22,22%     | 28,70%         |
| Insurance        | 18,81%      | 24,86%     | 22,71%      | 14,29%     | 26,22%         |
| Internet         | 22,05%      | 20,62%     | 24,20%      | 23,08%     | 22,17%         |
| TV               | 22,71%      | 23,73%     | 19,90%      | 44,44%     | 21,74%         |
| Water            | 20,95%      | 24,12%     | 25,50%      | 12,50%     | 21,33%         |

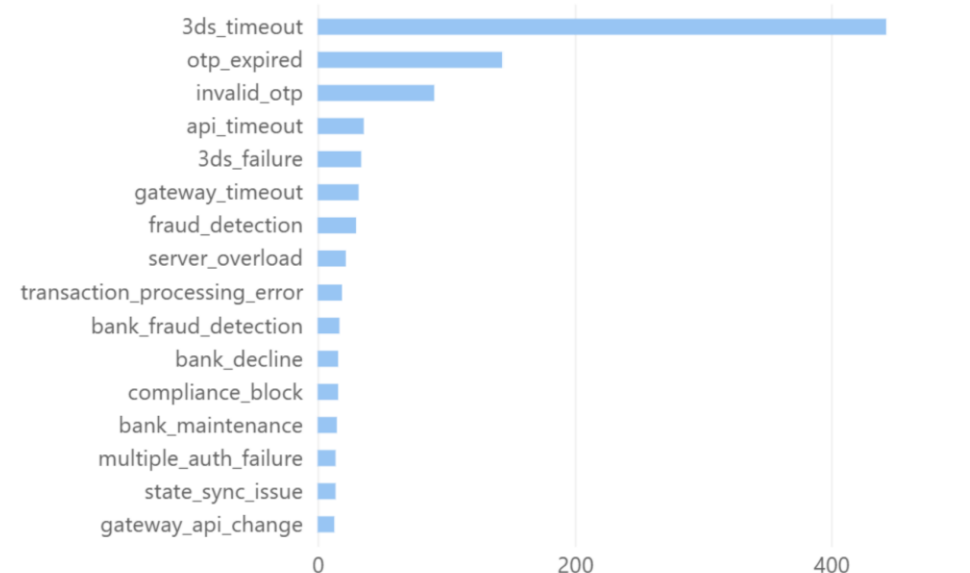
Failure rates are distributed evenly across standard payment methods (Credit Card, Debit Card, Linked Bank, Wallet Balance), ranging tightly from 17% to 28%. Local Card fluctuations stem entirely from statistical noise due to small sample sizes.

## 57.7% of Billing errors get stuck at the Authentication step

Total transaction failure by error\_category\_l2



Total transaction failure by error\_reason



While the system tags 63.5% of errors as user factors, L2 drill-downs reveal Authentication issues account for an overwhelming 57.7%. The destructive trio includes 3ds\_timeout (60.2%), otp\_expired (19.6%), and invalid\_otp (12.4%).

# Timeout expansion trade-offs and underlying memory leak risks

| app_version_group           | 3ds_timeout | api_timeout | gateway_timeout | otp_expired |
|-----------------------------|-------------|-------------|-----------------|-------------|
| 3.9.0 – 3.9.5 (before 28/7) | 0,93%       | 0,19%       | 0,20%           | 0,29%       |
| 3.9.6 (28/7)                | 2,77%       | 0,23%       | 0,18%           | 0,96%       |
| 3.9.7 (5/8)                 | 1,58%       | 0,40%       | 0,25%           | 0,46%       |
| 3.9.8 (15/8)                | 1,06%       | 0,31%       | 0,31%           | 0,82%       |
| 3.9.9 (emergency 22/8)      | 2,00%       | 0,18%       | 0,05%           | 0,59%       |

Relaxing user timeouts reduced 3DS/OTP errors, but migrating to Payment Gateway API v3.0 doubled Backend errors. Emergency update v3.9.9 lowered Bank gateway timeouts but left Frontend user timeouts unextended, causing 3ds\_timeouts to surge back. Unreleased memory combined with heavy 3DS webviews likely causes background crashes.

# Technical & Experience Optimization Proposals

## Memory Reconfiguration

Rollback or fine-tune the "Decreased memory usage" feature introduced in v3.9.6. Allocate sufficient RAM for payment workflows to prevent OS background termination.

## Session Recovery

Implement a state preservation mechanism. If apps reload when users switch out for OTP retrieval, save states so users can resume smoothly without timeout errors.

## Frontend Timeout Sync

Synchronize OTP/3DS frontend timeouts with full session lifespans, rather than solely relaxing API connection timeouts toward the backend/gateway like in v3.9.9.